

DINOSAUR 2 E, CO, USA

WMO: 745190

Lat: 40.2446N

Lon: 108.9677W

Elev: 1848

StdP: 81.01

Time Zone: -7.00 (NAM)

Period: 04-19

WBAN: 94082

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.5	-16.2	-22.5	0.6	-13.8	-20.2	0.8	-13.0	5.6	-5.8	4.5	-5.0	1.1	N/A	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	33.9	13.2	32.6	13.0	31.3	12.9	15.8	25.9	15.2	25.8	14.7	25.5	3.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.3	12.0	17.7	12.0	11.0	17.8	10.7	10.0	18.1	51.2	25.6	49.2	25.8	47.3	25.5	18.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.6	4.8	4.0	-21.7	36.5	2.7	0.8	-23.6	37.1	-25.2	37.6	-26.7	38.1	-28.7	38.7		
			-22.4	17.1	3.1	0.6	-24.6	17.5	-26.4	17.9	-28.2	18.3	-30.4	18.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.4	-5.7	-1.8	4.3	8.3	13.6	20.7	24.5	22.5	17.7	9.5	2.9	-4.8
	DBStd	11.19	5.55	5.35	4.78	4.66	4.71	4.39	2.45	2.65	4.33	4.86	4.95	5.59
	HDD10.0	1857	487	332	186	86	20	1	0	0	4	66	216	459
	HDD18.3	3755	746	565	435	302	158	28	0	4	62	273	464	718
	CDD10.0	1623	0	0	9	35	131	323	450	388	234	52	2	0
	CDD18.3	479	0	0	0	1	11	100	192	133	42	1	0	0
	CDH23.3	5162	0	0	0	8	142	1200	2083	1302	418	8	0	0
	CDH26.7	2138	0	0	0	1	25	505	983	513	110	1	0	0
(14) Wind	WSAvg	2.1	1.4	1.8	2.2	2.5	2.5	2.5	2.3	2.3	2.2	2.0	1.7	1.5
(15) Precipitation	PrecAvg	288	17	17	21	29	29	21	19	25	36	38	21	14
	PrecMax	389	37	29	66	77	73	88	51	67	99	99	46	73
	PrecMin	181	2	0	3	7	1	3	1	5	6	2	5	3
	PrecStd	57	10	8	14	17	20	22	14	14	22	24	12	15
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.7	12.9	20.1	24.3	29.6	35.1	36.1	33.9	31.8	24.4	16.4	9.5
		MCWB	1.5	4.4	7.5	9.3	11.1	13.1	14.4	13.3	12.6	10.0	6.6	3.1
	2%	DB	4.6	9.8	17.1	21.4	26.9	33.2	34.4	32.5	29.5	21.4	14.0	6.8
		MCWB	0.0	3.3	6.1	7.8	10.0	12.7	13.6	13.1	12.2	8.9	5.8	1.8
	5%	DB	2.9	7.4	14.6	19.3	24.9	31.7	33.1	31.2	27.8	19.1	12.1	4.8
		MCWB	-0.5	2.1	5.0	7.2	9.5	12.1	13.3	13.0	11.7	8.5	4.6	0.7
	10%	DB	1.4	5.3	12.2	16.9	22.4	29.8	31.9	29.9	25.9	17.0	10.2	2.8
		MCWB	-1.4	1.2	3.8	6.2	8.9	11.5	13.5	13.0	11.0	7.6	3.7	-0.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	5.4	8.1	9.7	12.0	14.8	16.9	16.3	15.4	12.3	7.5	4.1
		MCDB	5.6	11.8	19.7	22.0	24.1	28.9	26.7	27.4	25.0	19.1	15.0	8.3
	2%	WB	1.1	4.0	6.7	8.7	10.8	14.0	16.2	15.5	14.1	10.7	6.2	2.5
		MCDB	3.9	9.2	17.0	19.6	23.1	29.1	25.5	25.9	23.0	18.3	13.0	6.1
	5%	WB	0.0	2.8	5.5	7.8	10.1	13.2	15.6	15.0	13.4	9.7	5.1	1.4
		MCDB	2.6	7.6	14.1	17.5	21.8	28.2	25.9	25.1	22.2	17.2	11.0	4.7
	10%	WB	-1.1	1.7	4.4	6.7	9.4	12.4	15.0	14.3	12.6	8.6	3.9	-0.1
		MCDB	1.2	6.0	12.1	15.7	20.3	27.4	26.0	24.5	21.6	16.1	9.5	2.8
(35) Mean Daily Temperature Range	MDBR		7.7	8.7	10.7	12.4	13.1	15.0	14.7	14.3	13.4	11.1	9.3	7.6
	5% DB	MCDBR	8.8	10.3	13.5	15.3	15.7	16.4	15.9	15.6	15.1	13.3	11.3	9.1
		MCWBR	6.5	5.9	6.6	7.1	6.1	5.8	5.2	5.2	5.6	6.1	5.8	5.8
	5% WB	MCDBR	8.3	9.7	12.8	14.2	14.4	14.7	13.8	13.6	12.3	11.3	10.9	8.5
(40) Clear-Sky Solar Irradiance	taub		0.242	0.250	0.269	0.287	0.300	0.302	0.334	0.337	0.293	0.270	0.252	0.242
	taud		2.584	2.586	2.548	2.508	2.523	2.527	2.471	2.455	2.574	2.615	2.600	2.586
	Ebn at Noon		956	991	997	989	976	970	935	925	955	952	934	929
	Edh at Noon		67	78	91	102	103	103	108	107	88	75	66	62
(44) All-Sky Solar Radiation	RadAvg		2.30	3.20	4.57	5.79	6.76	7.80	7.14	6.35	5.42	3.87	2.64	1.98
	RadStd		0.18	0.23	0.33	0.33	0.41	0.45	0.32	0.26	0.32	0.27	0.20	0.17

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page